Felicity tech tree:
low temperature electrolysis of mixed metal oxides

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Abstract

Math behind the process for extracting metal from asteroid regolith with low temperature electrolysis.

Largely based on my interpretation of the paper "Low Temperature Electrolysis of Mixed Metal Oxides" by Everett Baker, Shaun Bonefas, Brittany Kyer, Christian Morneau, Yan Wang and Shawn Burdette

The final findings were that the machine included in the Felicity tech tree, which is used for this process, would require 348MJ of energy and 23.38h of time to process one 100-kilogram batch of asteroid regolith into mixed metal alloy. The energy cost of the alloy was found to be 14.7625MJ/kg, and a single machine allows for alloy production rate of 1.008246729 kg/h.

Contents

1 Basic idea of the process

Finely powdered asteroid regolith is added to an aqueous solution of sodium hydroxide, at concentrations of 50% to 60% by mass. Current is passed through the solution at 1.7 volts to reduce the metal oxides into metals, and deposit them on the cathode. This takes about 23.38 hours to complete.

Once the process is done, the cathode is removed with the metals crystallized onto it, which are removed for further processing.

Only a select group of elements whose oxides break down at or below the voltage used can be produced this way, so other elements will remain in oxidized state and have to be extracted in some other way.

What I actually sourced from the study:

- The material to extract metal from needs to be powdered - that is the form in which material was processed in the study.
- A solution which is 50% to 60% sodium hydroxide by mass with water.
- A current is passed through the solution at 1.7 volts.
- The exact list of elements which can be extracted using this method - see figure 2 in the original study (on page 9).

Parts of this which were my own additions:

- The extracted metals will crystallize onto the cathode. I've seen this process many times whenever passing current through a solution results in formation of solid material, but that's about all that my reasoning for stating that this would happen is based on.
- The crystallized metals will deposit as a mixed alloy rather than separate layers. This is a conservative assumption as on the contrary assuming that the metals will deposit in layers at this stage would trivialize their later separation in a way that gives overly optimistic energy costs.
- 23.38 hour time to completion - a successful sample took 5 hours to process in the study, but my calculations later down in the document showed that with the way and scale that I want to conduct the process at, thermal limitations increase the time to that much.

2 List of elements extracted via LTE

The listed elements are those which both can be extracted using Low Temperature Electrolysis and are present in asteroid regolith. For reasons I'm about to explain, "asteroid regolith" and "meteor" had been used interchangably here despite being obviously different, as that is an equivalence that my choice of my sources forced me to assume.

Ptable is an online interactive table of elements, which among other functions, provides the exact concentration of the chosen element as a percentage of mass in the whole universe, the solar system, meteor material, earth's crust, the ocean, and the human body. I chose the concentration in meteor material as likely the closest available approximation to asteroid material. Data was collected in 20.03.2025 and I had not re-checked it for any changes that might have happened since.

Atomic number	Symbol	Name	Percentage	Atomic mass	mass in 100kg
8	O	Oxygen	40	15.999	40kg (9.46kg)
26	Fe	Iron	22	55.845	22kg
27	Co	Cobalt	0.059	58.93319	59g
28	Ni	Nickel	1.3	58.6934	1.3kg
29	Cu	Copper	0.011	63.546	11g
30	Zn	Zinc	0.018	65.38	18g
31	Ga	Gallium	0.00076	69.723	760mg
32	Ge	Germanium	0.0021	72.63	2.1g
33	As	Arsenic	0.00018	74.92160	180mg
34	Se	Selenium	0.0013	78.971	1.3g
42	Mo	Molybdenum	0.00012	95.95	120mg
44	Ru	Ruthenium	0.000081	101.07	81mg
45	Rh	Rhodium	0.000018	102.9055	16mg
46	Pd	Palladium	0.000066	106.42	66mg
47	Ag	Silver	0.000014	107.8682	14mg
48	Cd	Cadmium	0.000044	112.414	44mg
49	In	Indium	0.0000044	114.818	4.4mg
50	Sn	Tin	0.00012	118.710	120mg
51	Sb	Antimony	0.000012	121.760	12mg
52	Te	Tellurium	0.00021	127.60	210mg
74	W	Tungsten	0.000012	183.84	12mg
75	Re	Rhenium	0.000000049	186.207	4.9mg
76	Os	Osmium	0.000066	190.23	66mg
77	Ir	Iridium	0.000054	192.217	54mg
78	Pt	Platinum	0.000098	195.084	98mg
79	Au	Gold	0.000017	196.9666	17mg
80	Hg	Mercury	0.000025	200.59	25mg
81	Tl	Thallium	0.0000078	204.38	7.8mg
82	Pb	Lead	0.00014	207.2	140mg
83	Bi	Bismuth	0.0000069	208.9804	6.9mg

The expected yield from processing 100kg of dry asteroid regolith is **23.573279kg** of metal and 9.46kg of oxygen. (the 40kg figure is how much total oxygen there is in 100kg of regolith, only 9.46kg actually gets released) There will remain a sludge including 66.966721 kg of the remaining elements which could not be brought to their elemental, still oxidized and mixed with the sodium hydroxide solution whose mass had not been counted. This byproduct can be simply washed to recover and reuse the sodium hydroxide.

3 Energy use & oxygen release

The study presents 2 chemical processes that happen during the electrolysis process:

$$Fe_2O_{3(s)} + 6e^- \rightarrow 2Fe_{(s)} + {}^3/_2O_{2(g)} + 6e^-$$

$$NiFe_2O_{4(s)}8e^- \rightarrow Ni_{(s)} + 2Fe_{(s)} + 4O_{2(s)} + 8e^-$$

Of those Fe_2O_3 (Iron(III) oxide) has a much more easily available enthalpy of formation, so it will be used to judge the energy cost, as well as the amount of oxygen released. These will necessarily only be approximations.

Iron has a molar mass of 0.0558kg/mol, and Nickel has a molar mass of 0.05869kg/mol.

$$\frac{22kg}{0.0558kg/mol} = 394mol$$

394 moles of atomic iron are produced. As Iron(III) oxide has 2 atoms of iron in it, that would mean each mole of Iron(III) oxide broken down would be responsible for 2 moles of iron extracted. 197 moles of Iron(III) oxide have to be broken down to yield 394 moles of iron. Doing so would also release 591 moles of oxygen atoms, equal to 9.46kg of oxygen.

The enthalpy of formation of Iron(III) oxide is -824.2 kJ/mol, which as I understand means that Fe_2O_3 releases 824.2 kJ/mol when formed from single free atoms of iron and oxygen. Using gibbs free energy might be more appropriate, which would be -742.2 kJ/mol, but I honestly don't know. Using the enthalpy of formation, breaking down all 197 moles of Iron(III) oxide would consume 162.3674MJ of energy:

$$824.2kJ/mol \cdot 197mol = 162.3674MJ$$

Now, that only accounts for the combined 22kg of iron produced, whereas the total mass of extracted metal is a bit over 23.57kg, and also assumes 100% efficiency. Majority of the remaining elements are heavier, and as such their molar fraction would be even smaller than their mass fraction, but as I had not done the calculations behind that, the mass fraction will do as a conservative estimate.

Additionally, I will assume a 50% efficiency, which feels reasonable for electrolysis, as the electrolysis of water typically has an efficiency well above that, though admittedly it is far from being the same thing.

$$162.3674MJ \cdot \frac{23.573279kg}{22kg} \cdot \frac{1}{0.5} = 347.9574564MJ$$

So the energy cost is **348 MJ** of electrical power for every 100kg batch of regolith.

From another perspective, the cost of the metal alloy obtained this way is roughly **14.7625MJ/kg**.

$$\frac{348MJ}{23.573279kg} = 14.7625MJ/kg$$

4 Power use

In the study, successful electrolysis of a sample required 5 hours, which would mean that the electrolysis pulls just 19331 watts, a massive amount of power. The solution needs to be hot, around 100^{o}C, but not boil, and assuming the 50% efficiency, half the power is just turned into heat.

I have honestly no idea how to calculate convection cooling, so I will substitute with the usual stefan-boltzmann law, and write it off as a conservative assumption because radiative cooling would occur either way and is slower than convection or conduction.

$$\sigma \cdot \epsilon \cdot A \cdot T^4 = P$$

where:

- σ is the stefan-boltzmann constant ($\approx 0.000000056703744 19W \cdot m^{-2} \cdot K^{-4}$)
- ϵ is the emissivity of material used
- A is the surface area of the hot object
- T is the temperature of the hot object
- P is the power of thermal radiation, ie. rate at which the object's heat is lost to glowing at various wavelengths of light due to its heat

I usually compare the mixed metal alloy resulting from the process to wrought iron, for which the engineering toolbox gives an emissivity of 0.94. In any case it would not be very difficult to artificially increase the emissivity of the device's walls up to that value, so that's what I'll use. As a sidenote this might imply the device will need to have additional radiating surfaces, which is nice.

If the electrolysis chamber should hold 100kg of asteroid material then it will likely hold about 200kg of total mass, might be about 200L in size, and if it were a cube that would give it a surface area of about 2m^2.

Temperature should be around 100^{o}C which is 373.15K.

$$\sigma \cdot 0.94 \cdot 2m^2 \cdot (373.15K)^4 = 2066.823399W$$

So the device can handle 2067W of heat output, which accounts for only half the power consumption because the second half is absorbed by the breakage of chemical bonds through electrolysis and as such is not contributing to the heating, so really the device should consume 4134W of power.

$$\frac{347.9574564MJ}{4134W} = 84169.67983s \approx 23h + 22min + 50s$$

So the time required to proces a 100kg batch of regolith is **23.38h** or **84.17ks** of time.

A single machine can allow an alloy production rate of **1.008246729 kg/h**.

$$\frac{23.573279kg}{84169.67983s} = 1.008246729kg/h$$

An interesting bit of trivia I can share that I originally did a little napkin math only loosely based on my vague estimates of the amperage and time required, which yielded 140MJ per batch, which as it turns out, would have actually been really close if I had also tacked on the "50% efficiency" debuff onto that estimate. At least I assume that's how I calculated that placeholder figure, as it is napkin math that I never wrote down, and as such don't really know.